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Inventor(s)	Kaiduka; Atsushi et al.

Non-aqueous electrolyte secondary cell

Abstract

The non-aqueous electrolyte secondary cell according to an embodiment of the present disclosure has a positive electrode, a negative electrode, and a non-aqueous electrolytic solution. The negative electrode has a negative electrode collector and a negative electrode active material layer provided on the negative electrode collector. The negative electrode active material layer contains graphite particles A and graphite particles B as negative electrode active materials. The graphite particles A have an internal void rate of 5% or below. The graphite particles B have an internal void rate of 8 to 20%. When the negative electrode active material layer is halved in the thickness direction, a region on the half closer to the outer surface contains more graphite particles A than a region on the half closer to the negative electrode collector.

Inventors:	Kaiduka; Atsushi (Osaka, JP), Takai; Yasuyuki (Osaka, JP)
Applicant:	SANYO Electric Co., Ltd. (Kadoma, JP)
Family ID:	76222193
Assignee:	PANASONIC ENERGY CO., LTD. (Osaka, JP)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a National Phase Application filed under 35 U.S.C. § 371 of International Application No. PCT/JP2020/043707 filed on Nov. 25, 2020 which claims the benefit of priority under 35 U.S.C. § 119(a) of Japanese Patent Application No. 2019-221682 filed in Japan on Dec. 6, 2019, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a non-aqueous electrolyte secondary battery.

BACKGROUND

(3) A non-aqueous electrolyte secondary battery using a carbon material as a negative electrode active material is widely used as a secondary battery having a high energy density.

(4) Patent Literature 1, for example, discloses a non-aqueous electrolyte secondary battery using densified carbon having an internal porosity of 5% or less as a carbon material.

(5) Patent Literature 2, for example, discloses a non-aqueous electrolyte secondary battery using a carbon material containing a carbon material A having an internal porosity of 1% or more and less than 23% and a carbon material B having an internal porosity of 23% or more and 40% or less.

CITATION LIST

Patent Literature

(6) Patent Literature 1: JP H9-320600 A Patent Literature 2: JP 2014-67638 A

SUMMARY

Technical Problem

(7) An object of the present disclosure is to provide a non-aqueous electrolyte secondary battery capable of suppressing deterioration of high-temperature storage characteristics and low-temperature discharge characteristics.

Solution to Problem

(8) A non-aqueous electrolyte secondary battery according to an aspect of the present disclosure includes: a positive electrode; a negative electrode; and a non-aqueous electrolyte solution, wherein the negative electrode includes a negative electrode current collector and a negative electrode active material layer provided on the negative electrode current collector, the negative electrode active material layer contains graphite particles A and graphite particles B as a negative electrode active material, an internal porosity of the graphite particles A is 5% or less, and an internal porosity of the graphite particles B is 8% to 20%, the graphite particles A are contained more in a half region on an outer surface side of the negative electrode active material layer than in a half region on a negative electrode current collector side of the negative electrode active material layer when the negative electrode active material layer is divided into two equal parts in a thickness direction, and the non-aqueous electrolyte solution contains a Li imide salt.

Advantageous Effects of Invention

(9) According to an aspect of the present disclosure, it is possible to suppress deterioration of high-temperature storage characteristics and low-temperature discharge characteristics.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a cross-sectional view of a non-aqueous electrolyte secondary battery as an example of an embodiment.

(2) FIG. 2 is a cross-sectional view of a negative electrode as an example of an embodiment.

(3) FIG. 3 is a cross-sectional view of a graphite particle in a negative electrode active material layer.

DESCRIPTION OF EMBODIMENTS

(4) A non-aqueous electrolyte secondary battery according to an aspect of the present disclosure includes: a positive electrode; a negative electrode; and a non-aqueous electrolyte solution, wherein the negative electrode includes a negative electrode current collector and a negative electrode active material layer provided on the negative electrode current collector, the negative electrode active material layer contains graphite particles A and graphite particles B as a negative electrode active material, an internal porosity of the graphite particles A is 5% or less, and an internal porosity of the graphite particles B is 8% to 20%, the graphite particles A are contained more in a half region on an outer surface side of the negative electrode active material layer than in a half region on a negative electrode current collector side of the negative electrode active material layer when the negative electrode active material layer is divided into two equal parts in a thickness direction, and the non-aqueous electrolyte solution contains a Li imide salt.

(5) It is considered that as in the present disclosure, a non-aqueous electrolyte solution containing a Li imide salt has higher thermal stability than that of a non-aqueous electrolyte solution containing no Li imide salt, so that the decomposition reaction of the non-aqueous electrolyte solution in a high-temperature environment is suppressed. As a result, deterioration of high-temperature storage characteristics of the non-aqueous electrolyte secondary battery is suppressed. On the other hand, since the non-aqueous electrolyte solution containing a Li imide salt has a higher viscosity than that of the non-aqueous electrolyte solution containing no Li imide salt, the permeation of the non-aqueous electrolyte solution to the negative electrode active material layer tends to be deteriorated in a low-temperature environment. However, as in the present disclosure, the graphite particles A having an internal porosity of 5% or less are contained more in a half region on the outer surface side of the negative electrode active material layer than in a half region on the negative electrode current collector side of the negative electrode active material layer, whereby a gap serving as a flow path of the non-aqueous electrolyte solution is secured from the outer surface to the inside of the negative electrode active material layer. As a result, deterioration of the permeation of the non-aqueous electrolyte solution to the negative electrode active material layer is suppressed even in a low-temperature environment, and deterioration of low-temperature discharge characteristics of the non-aqueous electrolyte secondary battery is suppressed.

(6) Hereinafter, an example of an embodiment will be described in detail with reference to the drawings. The non-aqueous electrolyte secondary battery of the present disclosure is not limited to the embodiments described below. The drawings referred to in the description of the embodiments are schematically illustrated.

(7) FIG. 1 is a cross-sectional view of a non-aqueous electrolyte secondary battery as an example of an embodiment. A non-aqueous electrolyte secondary battery **10** illustrated in FIG. 1 includes a wound electrode assembly **14** formed by winding a positive electrode **11** and a negative electrode **12** with a separator **13** interposed therebetween, a non-aqueous electrolyte solution, insulating plates **18** and **19** respectively disposed above and below the electrode assembly **14**, and a battery case **15** housing the above-described members. The battery case **15** includes a bottomed cylindrical case body **16** and a sealing assembly **17** that closes an opening of the case body **16**. Instead of the wound electrode assembly **14**, another form of electrode assembly such as a stacked electrode assembly in which a positive electrode and a negative electrode are alternately stacked with a separator interposed therebetween may be applied. Examples of the battery case **15** include metal

exterior cans having a cylindrical shape, a rectangular shape, a coin shape, a button shape, or the like, and a pouch exterior body formed by laminating a resin sheet and a metal sheet.

(8) The case body **16** is, for example, a bottomed cylindrical metal exterior can. A gasket **28** is provided between the case body **16** and the sealing assembly **17** to secure the sealing property of the inside of the battery. The case body **16** has, for example, a protruding portion **22** which supports the sealing assembly **17** and in which a part of a side surface portion extends inward. The protruding portion **22** is preferably formed in an annular shape along the circumferential direction of the case body **16**, and supports the sealing assembly **17** on the upper surface thereof.

(9) The sealing assembly **17** has a structure in which a filter **23**, a lower valve member **24**, an insulating member **25**, an upper valve member **26**, and a cap **27** are stacked in this order from the electrode assembly **14** side. Each member constituting the sealing assembly **17** has, for example, a disk shape or a ring shape, and the members excluding the insulating member **25** are electrically connected to each other. The lower valve member **24** and the upper valve member **26** are connected to each other at each of the central portions thereof, and the insulating member **25** is interposed between the peripheral portions of these valve bodies. When the internal pressure of the non-aqueous electrolyte secondary battery **10** increases due to heat generation caused by an internal short circuit or the like, for example, the lower valve member **24** deforms and breaks so as to push up the upper valve member **26** toward the cap **27**, and the current path between the lower valve member **24** and the upper valve member **26** is cut off. When the internal pressure further increases, the upper valve member **26** breaks, and the gas is discharged from the opening of the cap **27**.

(10) In the non-aqueous electrolyte secondary battery **10** illustrated in FIG. **1**, a positive electrode lead **20** attached to the positive electrode **11** extends to the sealing assembly **17** side through a through hole of the insulating plate **18**, and a negative electrode lead **21** attached to the negative electrode **12** extends toward the bottom side of the case body **16** through the outside of the insulating plate **19**. The positive electrode lead **20** is connected to the lower surface of the filter **23** which is the bottom plate of the sealing assembly **17** by welding or the like, and the cap **27** which is the top plate of the sealing assembly **17** electrically connected to the filter **23** serves as a positive electrode terminal. The negative electrode lead **21** is connected to the inner surface of the bottom portion of the case body **16** by welding or the like, and the case body **16** serves as a negative electrode terminal.

(11) Hereinafter, each component of the non-aqueous electrolyte secondary battery **10** will be described in detail.

(12) [Negative Electrode]

(13) FIG. **2** is a cross-sectional view of a negative electrode as an example of an embodiment. The negative electrode **12** includes a negative electrode current collector **40** and a negative electrode active material layer **42** provided on the negative electrode current collector **40**.

(14) As the negative electrode current collector **40**, for example, a foil of a metal which is stable in a potential range of the negative electrode, such as copper, a film in which the metal is disposed on a surface layer thereof, or the like is used.

(15) The negative electrode active material layer **42** contains graphite particles as a negative electrode active material. The negative electrode active material layer **42** preferably contains a binder or the like. The negative electrode **12** can be produced, for example, by preparing a negative electrode mixture slurry containing a negative electrode active material, a binder, and the like, applying the negative electrode mixture slurry onto the negative electrode current collector **40**, drying the slurry to form the negative electrode active material layer **42**, and rolling the negative electrode active material layer **42**. The method for producing the negative electrode active material layer **42** will be described later in detail.

(16) FIG. **3** is a cross-sectional view of a graphite particle in a negative electrode active material layer. As illustrated in FIG. **3**, a graphite particle **30** has a closed void **34** (hereinafter, internal void **34**) that is not connected from the inside of the particle to the surface of the particle and a void **36**

(hereinafter, external void **36**) that is connected from the inside of the particle to the surface of the particle in the cross-sectional view of the graphite particle **30**.

(17) The graphite particle **30** in the present embodiment includes a graphite particle A having an internal porosity of 5% or less and a graphite particle B having an internal porosity of 8% to 20%. The internal porosity of the graphite particles A may be 5% or less, but is preferably 1% to 5%, and more preferably 3% to 5%, from the viewpoint of suppressing deterioration of low-temperature discharge characteristics, and the like. The internal porosity of the graphite particles B may be 8% to 20%, and is preferably 10% to 18%, and more preferably 12% to 16%, from the viewpoint of suppressing deterioration of low-temperature discharge characteristics, and the like. Here, the internal porosity of the graphite particle is a two-dimensional value obtained from the ratio of the area of the internal void **34** of the graphite particle to the cross-sectional area of the graphite particle. The internal porosity of the graphite particle is determined by the following procedure.

(18) <Method for Measuring Internal Porosity>

(19) (1) The cross section of the negative electrode active material layer is exposed. Examples of the method for exposing the cross section include a method in which a part of the negative electrode is cut out and processed by an ion milling apparatus (for example, IM 4000 PLUS, manufactured by Hitachi High-Tech Corporation) to expose the cross section of the negative electrode active material layer. (2) A reflected electron image of the cross section of the exposed negative electrode active material layer is taken using a scanning electron microscope. The magnification at the time of taking the reflected electron image is 3,000 times to 5,000 times. (3) The cross-sectional image obtained as described above is taken into a computer, and binarized using image analysis software (for example, ImageJ, manufactured by National Institutes of Health, USA) to obtain a binarized image in which the particle cross section in the cross-sectional image is converted into black and voids present in the particle cross section are converted into white. (4) Graphite particles A and B having a particle size of 5 μm to 50 μm are selected from the binarized image, and the area of the cross section of the graphite particle and the area of internal voids present in the cross section of the graphite particle are calculated. Here, the area of the cross section of the graphite particle refers to the area of a region surrounded by the outer periphery of the graphite particle, that is, the area of the entire cross section portion of the graphite particle. In addition, for voids having a width of 3 μm or less among voids present in the cross section of the graphite particle, it may be difficult to determine whether the voids are internal voids or external voids in image analysis, and thus voids having a width of 3 μm or less may be considered to be internal voids. Then, the internal porosity of the graphite particle (the area of the internal voids of the cross section of the graphite particle $\times 100$ / the area of the cross section of the graphite particle) is calculated from the calculated area of the cross section of the graphite particle and area of the internal voids of the cross section of the graphite particle. The internal porosities of the graphite particles A and B are an average value of 10 graphite particles A and graphite particles B.

(20) The graphite particles A and B are produced, for example, as follows.

(21) <Graphite Particles A Having Internal Porosity of 5% or Less>

(22) For example, coke (precursor) as a main raw material is pulverized into a predetermined size, the pulverized coke is aggregated with a binder, and then the coke is fired at a temperature of 2,600° C. or higher to be graphitized and sieved to obtain graphite particles A having a desired size. Here, the internal porosity can be adjusted to 5% or less depending on the particle size of the precursor after pulverization, the particle size of the precursor in an aggregated state, and the like. The average particle size (volume-based median diameter D50) of the precursor after pulverization is preferably in the range of 12 μm to 20 μm , for example. When the internal porosity is reduced to 5% or less, it is preferable to increase the particle size of the precursor after pulverization.

(23) <Graphite Particles B Having Internal Porosity of 8% to 20%>

(24) For example, coke (precursor) as a main raw material is pulverized into a predetermined size, the pulverized coke is aggregated with a binder, and then the coke is further pressure-formed into a

block shape and then fired at a temperature of 2,600° C. or higher to be graphitized. The block-shaped formed body after graphitization is pulverized and sieved to obtain graphite particles B having a desired size. Here, the internal porosity can be adjusted to 8% to 20% depending on the amount of the volatile component added to the block-shaped formed body. When a part of the binder added to the coke (precursor) volatilizes during firing, the binder can be used as a volatile component. As such a binder, pitch is exemplified.

(25) The graphite particles A and B used in the present embodiment are natural graphite, artificial graphite, and the like, and are not particularly limited, but artificial graphite is preferable from the viewpoint of ease of adjusting the internal porosity and the like. The interplanar spacing

(d.sub.002) of the (002) plane of the graphite particles A and B used in the present embodiment as measured by a wide angle X-ray diffraction method is, for example, preferably 0.3354 nm or more, more preferably 0.3357 nm or more, and preferably less than 0.340 nm, more preferably 0.338 nm or less. In addition, the crystallite size (Lc (002)) of the graphite particles A and B used in the present embodiment determined by an X-ray diffraction method is, for example, preferably 5 μm or more, more preferably 10 nm or more, and preferably 300 nm or less, more preferably 200 nm or less. When the interplanar spacing (d.sub.002) and the crystallite size (Lc (002)) satisfy the above ranges, the battery capacity of the non-aqueous electrolyte secondary battery tends to be larger than the case where the above ranges are not satisfied.

(26) In the present embodiment, the graphite particles A are contained more in a region **42b** which is a half region on the outer surface side than in a region **42a** which is a half region on the negative electrode current collector side when the negative electrode active material layer **42** illustrated in FIG. 2 is divided into two equal parts in the thickness direction. As a result, deterioration of the permeation of the non-aqueous electrolyte solution to the negative electrode active material layer **42** is suppressed even in a low-temperature environment, so that deterioration of low-temperature discharge characteristics of the non-aqueous electrolyte secondary battery is suppressed. The phrase “the negative electrode active material layer **42** is divided into two equal parts in the thickness direction” means that when the stacking direction of the negative electrode current collector **40** and the negative electrode active material layer **42** is defined as the thickness direction of the negative electrode active material layer **42**, the negative electrode active material layer **42** is divided into two equal parts at an intermediate Z of the thickness of the negative electrode active material layer **42**. Then, among the two equal parts divided in the thickness direction of the negative electrode active material layer **42**, a part of the negative electrode active material layer **42** located close to the negative electrode current collector **40** is defined as the region **42a** which is a half region on the negative electrode current collector side, and a part of the negative electrode active material layer **42** located far from the negative electrode current collector **40** is defined as a region **42b** which is a half region on the outer surface side.

(27) In the present embodiment, the graphite particles A may be contained more in the region **42b** which is a half region on the outer surface side than in the region **42a** which is a half region on the negative electrode current collector side, and from the viewpoint of suppressing deterioration of low-temperature discharge characteristics, the ratio of the graphite particles A to the graphite particles B in the region **42b** which is a half region on the outer surface is preferably 20:80 to 100:0, and more preferably 50:50 to 100:0 in mass ratio. Furthermore, the ratio of the graphite particles A to the graphite particles B in the region **42a** which is a half region on the negative electrode current collector side is preferably 10:90 to 0:100, and more preferably 0:100 in mass ratio.

(28) An example of a method for producing the negative electrode active material layer **42** will be described. For example, a negative electrode active material containing graphite particles B (as necessary, graphite particles A), a binder, and a solvent such as water are mixed to prepare a negative electrode mixture slurry for the negative electrode current collector side. Separately from this, a negative electrode active material containing graphite particles A (as necessary, graphite

particles B) in an amount larger than that of the negative electrode mixture slurry for the negative electrode current collector side, a binder, and a solvent such as water are mixed to prepare a negative electrode mixture slurry for the outer surface side. Then, the negative electrode mixture slurry for the negative electrode current collector side is applied onto both surfaces of the negative electrode current collector and dried, and then the negative electrode mixture slurry for the outer surface side is applied onto both surfaces of the coating film of the negative electrode mixture shiny for the negative electrode current collector side and dried, whereby the negative electrode active material layer **42** can be formed. In the above method, the negative electrode mixture slurry for the negative electrode current collector side is applied and dried, and then the negative electrode mixture slurry for the outer surface side is applied, but a method of applying the negative electrode mixture slurry for the outer surface side after applying the negative electrode mixture slurry for the negative electrode current collector side and before drying may be used, or the negative electrode mixture slurry for the negative electrode current collector side and the negative electrode mixture shiny for the outer surface side may be applied simultaneously.

(29) The negative electrode active material may contain other materials capable of reversibly absorbing and releasing lithium ions in addition to the graphite particles A and B used in the present embodiment, and may contain, for example, a Si-based material. Examples of the Si-based material include Si, alloys containing Si, and silicon oxides such as SiO_x (x is 0.8 to 1.6). The Si-based material is a negative electrode material capable of improving the battery capacity as compared with the graphite particles. On the other hand, the Si-based material is disadvantageous in terms of charge-discharge cycle characteristics because the volume expansion of the Si-based material associated with charge and discharge is large. However, in the negative electrode active material layer having a negative electrode active material containing the graphite particles A and B and the Si-based material, when the amount of the graphite particles A is increased in the half region on the outer surface side rather than the half region on the negative electrode current collector side, deterioration of charge-discharge cycle characteristics may be effectively suppressed as compared with the case where the graphite particles A are uniformly dispersed in the negative electrode active material layer. The content of the Si-based material is, for example, preferably 1 mass % to 10 mass %, and more preferably 3 mass % to 7 mass %, with respect to the mass of the negative electrode active material, from the viewpoint of improving the battery capacity, suppressing deterioration of charge-discharge cycle characteristics, and the like.

(30) Other examples of the other materials capable of reversibly absorbing and releasing lithium ions include metals alloyed with lithium, such as tin (Sn), and alloys and oxides containing metal elements such as Sn. The negative electrode active material may contain the other materials, and the content of the other material is desirably, for example, 10 mass % or less with respect to the mass of the negative electrode active material.

(31) Examples of the binder include a fluorine resin, PAN, a polyimide resin, an acrylic resin, a polyolefin resin, styrene-butadiene rubber (SBR), nitrile-butadiene rubber (NBR), carboxymethyl cellulose (CMC) or a salt thereof, polyacrylic acid (PAA) or a salt thereof (PAA-Na, PAA-K, or the like, and partially neutralized salts may be used), and polyvinyl alcohol (PVA). These may be used singly or in combination of two or more types thereof.

(32) [Positive Electrode]

(33) The positive electrode **11** includes, for example, a positive electrode current collector such as a metal foil and a positive electrode active material layer formed on the positive electrode current collector. As the positive electrode current collector, a foil of a metal which is stable in a potential range of the positive electrode, such as aluminum, a film in which the metal is disposed on a surface layer thereof, or the like can be used. The positive electrode active material layer contains, for example, a positive electrode active material, a binder, a conductive material, and the like.

(34) The positive electrode **11** can be produced, for example, by applying a positive electrode mixture slurry containing a positive electrode active material, a binder, a conductive material, and

the like onto a positive electrode current collector, drying the slurry to form a positive electrode active material layer, and then rolling the positive electrode active material layer.

(35) Examples of the positive electrode active material include lithium transition metal oxides containing transition metal elements such as Co, Mn, and Ni. Examples of the lithium transition metal oxide include $\text{Li}_{0.5}\text{CoO}_{1.5}$, $\text{Li}_{0.5}\text{NiO}_{1.5}$, $\text{Li}_{0.5}\text{MnO}_{1.5}$, $\text{Li}_{0.5}\text{Co}_{0.5}\text{Ni}_{0.5}\text{O}_{1.5}$, $\text{Li}_{0.5}\text{Co}_{0.5}\text{M}_{0.5}\text{O}_{1.5}$, $\text{Li}_{0.5}\text{Ni}_{0.5}\text{M}_{0.5}\text{O}_{1.5}$, $\text{Li}_{0.5}\text{Mn}_{0.5}\text{O}_{1.5}$, $\text{Li}_{0.5}\text{Mn}_{0.5}\text{O}_{1.5}$, LiMPO_4 , and $\text{Li}_2\text{MPO}_4\text{F}$ (M; at least one of Na, Mg, Sc, Y, Mn, Fe, Co, Ni, Cu, Zn, Al, Cr, Pb, Sb, and B, $0 < x \leq 1.2$, $0 < y \leq 0.9$, $2.0 \leq z \leq 2.3$). These may be used singly or in combination of two or more types thereof. The positive electrode active material preferably contains a lithium-nickel composite oxide such as $\text{Li}_{0.5}\text{NiO}_{1.5}$, $\text{Li}_{0.5}\text{Co}_{0.5}\text{Ni}_{0.5}\text{O}_{1.5}$, or $\text{Li}_{0.5}\text{Ni}_{0.5}\text{M}_{0.5}\text{O}_{1.5}$ (M; at least one of Na, Mg, Sc, Y, Mn, Fe, Co, Ni, Cu, Zn, Al, Cr, Pb, Sb, and B, $0 < x \leq 1.2$, $0 < y \leq 0.9$, $2.0 \leq z \leq 2.3$) from the viewpoint of being able to increase the capacity of the non-aqueous electrolyte secondary battery.

(36) Examples of the conductive material include carbon particles such as carbon black (CB), acetylene black (AB), Ketjen black, and graphite. These may be used singly or in combination of two or more types thereof.

(37) Examples of the binder include fluorine resins such as polytetrafluoroethylene (PTFE) and polyvinylidene fluoride (PVdF), polyacrylonitrile (PAN), polyimide resins, acrylic resins, and polyolefin resins. These may be used singly or in combination of two or more types thereof.

(38) [Non-Aqueous Electrolyte Solution]

(39) The non-aqueous electrolyte solution contains a Li imide salt. As described above, since the non-aqueous electrolyte solution contains a Li imide salt, the thermal stability of the non-aqueous electrolyte solution is improved, so that the deterioration of high-temperature storage characteristics of the non-aqueous electrolyte secondary battery is suppressed.

(40) The Li imide salt is not particularly limited as long as it is a salt having a lithium ion and an imide anion, but for example, a Li imide salt represented by the following general formula is preferable.

$\text{Li}[\text{N}(\text{X}_1\text{SO})_2(\text{X}_2\text{SO})_2]$

(41) where X1 and X2 are each independently selected from a halogen atom, an alkyl group, and a halogen-substituted alkyl group. X1 and X2 may be bonded to each other to form a ring.

(42) The number of carbon atoms of the alkyl group and the halogen-substituted alkyl group is preferably 1 to 6, more preferably 1 to 3, and still more preferably 1 to 2. The halogen of the halogen atom or halogen-substituted alkyl group is preferably fluorine. The number of halogen substitutions in the halogen-substituted alkyl group is equal to or less than the number of hydrogens of the original alkyl group.

(43) Examples of the Li imide salt represented by the above general formula include lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), lithium bis(perfluoroethanesulfonyl)imide (LiBETI), lithium(perfluoroethanesulfonyl)(trifluoromethanesulfonyl)imide, lithium bis(fluorosulfonyl)imide (LiFSI), and lithium (fluorosulfonyl)(trifluoromethanesulfonyl)imide (LiFTI). Among them, lithium bis(fluorosulfonyl)imide (LiFSI) is preferable from the viewpoint of solubility in the non-aqueous electrolyte solution, thermal stability of the non-aqueous electrolyte solution, and the like. These may be used singly or in combination of two or more types thereof.

(44) The concentration of the Li imide salt in the non-aqueous electrolyte solution is, for example, preferably in the range of 0.05 mol/L to 1.5 mol/L, and more preferably in the range of 0.05 mol/L to 0.7 mol/L. When the concentration of the Li imide salt satisfies the above range, deterioration of the high-temperature storage characteristics of the non-aqueous electrolyte secondary battery may be further suppressed as compared with the case where the above range is not satisfied.

(45) The non-aqueous electrolyte solution may contain a Li salt other than the Li imide salt. Examples of the Li salt other than the Li imide salt include borates such as LiBF_4 ,

LiClO₄, LiPF₆, LiAsF₆, LiSbF₆, LiAlCl₄, LiSCN, LiCF₃SO₃, LiCF₃CO₂, Li(P(CO₂)₄F)₄, LiPF_{6-x}(C_nF_{2n+1})_x (1 < x < 6, n is 1 or 2), LiB₁₀Cl₁₀, LiCl, LiBr, LiI, chloroborane lithium, lithium lower aliphatic carboxylate, Li₂B₄O₇, and Li(B(CO₂)₄F)₂. These may be used singly or in combination of two or more types thereof. Among them, LiPF₆ is preferably used from the viewpoint of lithium ion conductivity, electrochemical stability, and the like. The concentration of the Li salt other than the Li imide salt in the non-aqueous electrolyte solution is preferably, for example, in the range of 0.8 mol/L to 1.8 mol.

(46) The non-aqueous electrolyte solution contains, for example, a non-aqueous solvent that dissolves an electrolyte salt such as a Li imide salt. As the non-aqueous solvent, for example, esters, ethers, nitriles such as acetonitrile, amides such as dimethylformamide, and mixed solvents of two or more types thereof can be used. The non-aqueous solvent may contain a halogen-substituted compound in which at least a part of hydrogen in these solvents is substituted with a halogen atom such as fluorine.

(47) Examples of the esters include cyclic carbonic acid esters such as ethylene carbonate (EC), propylene carbonate (PC), and butylene carbonate; chain carbonic acid esters such as dimethyl carbonate (DMC), ethyl methyl carbonate (EMC), diethyl carbonate (DEC), methyl propyl carbonate, ethyl propyl carbonate, and methyl isopropyl carbonate; cyclic carboxylic acid esters such as γ -butyrolactone and γ -valerolactone; and chain carboxylic acid esters such as methyl acetate, ethyl acetate, propyl acetate, methyl propionate (MP), and ethyl propionate.

(48) Examples of the ethers include cyclic ethers such as 1,3-dioxolane, 4-methyl-1,3-dioxolane, tetrahydrofuran, 2-methyltetrahydrofuran, propylene oxide, 1,2-butylene oxide, 1,3-dioxane, 1,4-dioxane, 1,3,5-trioxane, furan, 2-methylfuran, 1,8-cineol, and crown ether; and chain ethers such as 1,2-dimethoxyethane, diethyl ether, dipropyl ether, diisopropyl ether, dibutyl ether, dihexyl ether, ethyl vinyl ether, butyl vinyl ether, methyl phenyl ether, ethyl phenyl ether, butyl phenyl ether, pentyl phenyl ether, methoxytoluene, benzyl ethyl ether, diphenyl ether, dibenzyl ether, o-dimethoxybenzene, 1,2-diethoxy ethane, 1,2-dibutoxyethane, diethylene glycol dimethyl ether, diethylene glycol diethyl ether, diethylene glycol dibutyl ether, 1,1-dimethoxymethane, 1,1-diethoxyethane, triethylene glycol dimethyl ether, and tetraethylene glycol dimethyl ether.

(49) As the halogen-substituted compound, it is preferable to use a fluorinated cyclic carbonic acid ester such as fluoroethylene carbonate (FEC), a fluorinated chain carbonic acid ester, a fluorinated chain carboxylic acid ester such as methyl fluoropropionate (FMP), or the like.

(50) [Separator]

(51) As the separator **13**, for example, a porous sheet having ion permeability and insulating property is used. Specific examples of the porous sheet include a microporous thin film, a woven fabric, and a nonwoven fabric. As a material of the separator, an olefin resin such as polyethylene or polypropylene, cellulose, or the like is suitable. The separator **13** may be a laminate including a cellulose fiber layer and a thermoplastic resin fiber layer of an olefin resin or the like. In addition, a multilayer separator including a polyethylene layer and a polypropylene layer may be used, and a separator having a surface coated with a material such as an aramid resin or ceramic may be used.

EXAMPLES

(52) Hereinafter, the present disclosure will be further described with reference to examples, but the present disclosure is not limited to these examples.

Example 1

(53) [Production of Positive Electrode]

(54) As a positive electrode active material, aluminum-containing lithium nickel cobalt oxide (LiNi_{0.88}Co_{0.09}Al_{0.03}O₂) was used. First, 100 parts by mass of the positive electrode active material, 1 part by mass of acetylene black, and 0.9 parts by mass of polyvinylidene fluoride were mixed in a solvent of N-methyl-2-pyrrolidone (NMP) to prepare a positive electrode mixture slurry. This slurry was applied to both surfaces of an aluminum foil

having a thickness of 15 μm , the coating film was dried, and then the coating film was rolled by a rolling roller to produce a positive electrode in which a positive electrode active material layer was formed on both surfaces of a positive electrode current collector. The produced positive electrode was cut to a width of 57.6 mm and a length of 679 mm and used.

(55) [Production of Graphite Particles A]

(56) Coke was pulverized until the average particle size (median diameter D50) thereof reached 12 μm . Pitch as a binder was added to the pulverized coke, and the coke was aggregated until the average particle size (median diameter D50) thereof reached 17 μm . This aggregate was fired at a temperature of 2,800° C. to be graphitized, and then sieved using a 250 mesh sieve to obtain graphite particles A having an average particle size (median diameter D50) of 23 μm .

(57) [Production of Graphite Particles B]

(58) Coke was pulverized until the average particle size (median diameter D50) thereof reached 15 μm , pitch as a binder was added to the pulverized coke to aggregate the coke, and then a block-shaped formed body having a density of 1.6 g/cm^{sup.3} to 1.9 g/cm^{sup.3} was produced at an isotropic pressure. The block-shaped formed body was fired at a temperature of 2,800° C. to be graphitized. Next, the graphitized block-shaped formed body was pulverized, and sieved using a 250 mesh sieve to obtain graphite particles B having an average particle size (median diameter D50) of 23 μm .

(59) [Production of Negative Electrode]

(60) First, 40 parts by mass of the graphite particles A, 55 parts by mass of the graphite particles B, and 5 parts by mass of SiO were mixed, and the resulting mixture was used as a negative electrode active material A contained in the half region on the outer surface side of the negative electrode active material layer. The negative electrode active material A, carboxymethyl cellulose (CMC), and styrene-butadiene copolymer rubber (SBR) were mixed at a mass ratio of 100:1:1 to prepare a negative electrode mixture slurry for the outer surface side. Further, 95 parts by mass of the graphite particles B and 5 parts by mass of SiO were mixed, and the resulting mixture was used as a negative electrode active material B contained in the half region on the negative electrode current collector side of the negative electrode active material layer. The negative electrode active material B, carboxymethyl cellulose (CMC), and styrene-butadiene copolymer rubber (SBR) were mixed at a mass ratio of 100:1:1 to prepare a negative electrode mixture slurry for the negative electrode current collector side.

(61) The negative electrode mixture slurry for the negative electrode current collector side was applied to both surfaces of a copper foil having a thickness of 8 μm , the coating film was dried, and then the negative electrode mixture slurry for the outer surface side was applied onto the coating film and dried, and the coating film was rolled by a rolling roller to produce a negative electrode in which a negative electrode active material layer was formed on both surfaces of the negative electrode current collector. That is, the mass ratio of the graphite particles A to the graphite particles B in the half region on the outer surface side of the negative electrode active material layer is 40:55, and the mass ratio of the graphite particles A to the graphite particles B in the half region on the negative electrode current collector side of the negative electrode active material layer is 0:100. In the produced negative electrode, the internal porosities of the graphite particles A and B were measured to be 3% and 15%, respectively. The produced negative electrode was cut to a width of 58.6 mm and a length of 662 mm and used.

Preparation of Non-Aqueous Electrolyte Solution

(62) In a non-aqueous solvent obtained by mixing ethylene carbonate (EC), methyl ethyl carbonate (MEC), and dimethyl carbonate (DMC) at a volume ratio of 20:5:75, LiPF_{sub.6} was dissolved at a concentration of 1.4 mol/L, and 3 mass % of vinylene carbonate and 0.1 mol/L of LiFSI were further added. This was used as a non-aqueous electrolyte solution.

(63) [Production of Non-Aqueous Electrolyte Secondary Battery]

(64) (1) A positive electrode lead made of aluminum was attached to the positive electrode current

collector, and a negative electrode lead made of nickel-copper-nickel was attached to the negative electrode current collector, and then the positive electrode and the negative electrode were wound with a separator made of polyethylene interposed therebetween to produce a wound electrode assembly. (2) Insulating plates were disposed above and below the electrode assembly, respectively, the negative electrode lead was welded to a case body, the positive electrode lead was welded to a sealing assembly, and the electrode assembly was housed in the case body. (3) The non-aqueous electrolyte was injected into the case body by a decompression method, the open end of the case body was crimped to the sealing assembly via a gasket. This was used as a non-aqueous electrolyte secondary battery.

Comparative Example 1

(65) First, 95 parts by mass of the graphite particles B and 5 parts by mass of SiO were mixed, and the resulting mixture was used as a negative electrode active material C contained in the entire region of the negative electrode active material layer. The negative electrode active material C, carboxymethyl cellulose (CMC), and styrene-butadiene copolymer rubber (SBR) were mixed at a mass ratio of 100:1:1 to prepare a negative electrode mixture slurry. The negative electrode mixture slurry was applied to both surfaces of a copper foil having a thickness of 8 μm , the coating film was dried, and then the coating film was rolled by a rolling roller to prepare a negative electrode in which a negative electrode active material layer was formed on both surfaces of a negative electrode current collector. That is, the mass ratio of the graphite particles A to the graphite particles B in the half region on the outer surface side of the negative electrode active material layer is 0:100, and the mass ratio of the graphite particles A to the graphite particles B in the half region on the negative electrode current collector side of the negative electrode active material layer is 0:100. The produced negative electrode was cut to a width of 58.6 mm and a length of 662 mm and used.

(66) In a non-aqueous solvent obtained by mixing ethylene carbonate (EC), methyl ethyl carbonate (MEC), and dimethyl carbonate (DMC) at a volume ratio of 20:5:75, LiPF₆ was dissolved at a concentration of 1.4 mol/L, and 3 mass % of vinylene carbonate was further added. This was used as a non-aqueous electrolyte solution. That is, in Comparative Example 1, a non-aqueous electrolyte solution containing no LiFSI was used.

(67) Using the negative electrode and the non-aqueous electrolyte solution, a non-aqueous electrolyte secondary battery was produced in the same manner as in Example 1.

Comparative Example 2

(68) A non-aqueous electrolyte secondary battery was produced in the same manner as in Example 1 except that the negative electrode produced in Comparative Example 1 was used.

Comparative Example 3

(69) A non-aqueous electrolyte secondary battery was produced in the same manner as in Example 1 except that the non-aqueous electrolyte solution prepared in Comparative Example 1 was used.

(70) [High-Temperature Storage Test]

(71) The non-aqueous electrolyte secondary batteries of Example 1 and Comparative Examples were charged up to 4.2 V at a constant current of 990 mA (0.3 hour rate) at an environmental temperature of 25° C., and then charged at a constant voltage of 4.2 V with a termination current of 66 mA. Thereafter, constant current discharge was performed up to 3.0 V at a constant current of 1,650 mA (0.5 hour rate), and the discharge capacity at this time was measured as the battery capacity before storage. Thereafter, constant current charge was performed up to 4.2 V at a constant current of 990 mA (0.3 hour rate), constant voltage charge was performed at a constant voltage of 4.2 V with a termination current of 66 mA, and then the battery was stored in a thermostatic bath at 80° C. for 3 days. The non-aqueous electrolyte secondary battery after storage was taken out from the thermostatic bath, left for 1 hour, and then constant current discharge was performed up to 3.0 V at a constant current of 1,650 mA (0.5 hour rate) at an environmental temperature of 25° C. The discharge capacity at this time was measured as the battery capacity after high-temperature storage.

Then, the capacity maintenance rate after high-temperature storage was determined from the following equation.

Capacity maintenance rate after high-temperature storage=(battery capacity after storage/battery capacity before storage)×100

[Low-Temperature Discharge Test]

(72) The non-aqueous electrolyte secondary batteries of Example 1 and Comparative Examples were charged up to 4.2 V at a constant current of 990 mA (0.3 hour rate) at an environmental temperature of 25° C., and then charged at a constant voltage of 4.2 V with a termination current of 66 mA. Thereafter, constant current discharge was performed up to 3.0 V at a constant current of 3,300 mA (1.0 hour rate), and the discharge capacity at this time was measured as the battery capacity at 25° C. Thereafter, constant current charge was performed up to 4.2 V at a constant current of 990 mA (0.3 hour rate) at an environmental temperature of 25° C., constant voltage charge was performed at a constant voltage of 4.2 V with a termination current of 66 mA, and then the battery was left in an environment of −10° C. for 2 hours. Then, constant current discharge was performed up to 3.0 V at a constant current of 3,300 mA (1.0 hour rate) at an environmental temperature of −10° C., and the discharge capacity at this time was measured as the battery capacity at −10° C. The capacity maintenance rate in low-temperature discharge was determined from the following equation.

Capacity maintenance rate in low-temperature discharge=(battery capacity at −10° C./battery capacity at 25° C.)×100

(73) Table 1 summarizes the results of the capacity maintenance rate after high-temperature storage and the capacity maintenance rate in low-temperature discharge in the non-aqueous electrolyte secondary batteries of Example 1 and Comparative Examples. Here, as the value of the capacity maintenance rate after high-temperature storage is high, the deterioration of high-temperature storage characteristics is suppressed, and as the value of the capacity maintenance rate in low-temperature discharge is high, the deterioration of low-temperature discharge characteristics is suppressed.

(74) TABLE-US-00001 TABLE 1 Mass ratio of graphite particles A to graphite particles B
Capacity maintenance rate Half region on negative After high- Low- Half region on electrode
current temperature temperature outer surface side collector side LiFSI storage discharge Example
1 A:B = 40:55 A:B = 0:100 Present 94.4% 92.6% Comparative A:B = 0:100 A:B = 0:100 Absent
93.9% 92.1% Example 1 Comparative A:B = 0:100 A:B = 0:100 Present 94.4% 91.6% Example 2
Comparative A:B = 40:55 A:B = 0:100 Absent 93.9% 92.6% Example 3

(75) As can be seen from Table 1, the capacity maintenance rate after high-temperature storage in Example 1 was higher than those in Comparative Examples 1 and 3, and the capacity maintenance rate in low-temperature discharge in Example 1 was higher than those in Comparative Examples 1 and 2. That is, according to Example 1, it is possible to suppress deterioration of high-temperature storage characteristics and low-temperature discharge characteristics.

REFERENCE SIGNS LIST

(76) **10** Non-aqueous electrolyte secondary battery **11** Positive electrode **12** Negative electrode **13** Separator **14** Electrode assembly **15** Battery case **16** Case body **17** Sealing assembly **18, 19** Insulating plate **20** Positive electrode lead **21** Negative electrode lead **22** Protruding portion **23** Filter **24** Lower valve member **25** Insulating member **26** Upper valve member **27** Cap **28** Gasket **30** Graphite particle **34** Internal void **36** External void **40** Negative electrode current collector **42** Negative electrode active material layer **42a** Half region on negative electrode current collector side **42b** Half region on outer surface side

Claims

1. A non-aqueous electrolyte secondary battery comprising: a positive electrode; a negative electrode; and a non-aqueous electrolyte solution, wherein the negative electrode includes a negative electrode current collector and a negative electrode active material layer provided on the negative electrode current collector, the negative electrode active material layer contains graphite particles A and graphite particles B as a negative electrode active material, an internal porosity of the graphite particles A is 5% or less, and an internal porosity of the graphite particles B is 12% to 16%, the graphite particles A are contained more in a half region on an outer surface side of the negative electrode active material layer than in a half region on a negative electrode current collector side of the negative electrode active material layer when the negative electrode active material layer is divided into two equal parts in a thickness direction, and the non-aqueous electrolyte solution contains a Li imide salt.
 2. The non-aqueous electrolyte secondary battery according to claim 1, wherein the Li imide salt contains lithium bis(fluorosulfonyl)imide.
 3. The non-aqueous electrolyte secondary battery according to claim 1, wherein the negative electrode active material contains a Si-based material.
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